NEUROSCIENCE

Thalamic opioids from POMC satiety neurons switch on sugar appetite

Marielle Minère[1,2], Hannah Wilhelms[1], Bojana Kuzmanovic[3], Sofia Lundh[4], Debora Fusca[5,6], Alina Claßen[1], Stav Shtiglitz[7], Yael Prilutski[7], Itay Talpir[7], Lin Tian[8], Brigitte Kieffer[9], Jon Davis[10], Peter Kloppenburg[5,6], Marc Tittgemeyer[3,5,11], Yoav Livneh[7], Henning Fenselau[1,2,5]*

High sugar–containing foods are readily consumed, even after meals and beyond fullness sensation (e.g., as desserts). Although reward-driven processing of palatable foods can promote overeating, the neurobiological mechanisms that underlie the selective appetite for sugar in states of satiety remain unclear. Hypothalamic pro-opiomelanocortin (POMC) neurons are principal regulators of satiety because they decrease food intake through excitatory melanocortin neuropeptides. We discovered that POMC neurons not only promote satiety in fed conditions but concomitantly switch on sugar appetite, which drives overconsumption. POMC neuron projections to the paraventricular thalamus selectively inhibited postsynaptic neurons through mu-opioid receptor signaling. This opioid circuit was strongly activated during sugar consumption, which was most notable in satiety states. Correspondingly, inhibiting its activity diminished high-sugar diet intake in sated mice.

Satiety, the sensation of being sated or full, is an important neurobiological process for the maintenance of a stable body weight. Satiety is triggered by caloric surplus and in response to the resolution of caloric deficiency, such as after eating a meal (*1*, *2*). However, although feeding behavior and overall food intake are attenuated in these states of satiety, they are directly associated with an increased desire to eat sweet, high sugar–containing foods. This paradoxical increase in appetite for sugar is most notable after meals, which accounts for the widespread consumption of desserts. Brain pathways that encode reward-driven behavior critically contribute to food intake regulation and can drive consumption of palatable food beyond homeostatic needs (*3*). However, activity manipulations have shown that these pathways also control food intake in states of energy deprivation as well as the consumption of fat (*4*, *5*). Thus, despite considerable interest for the development of new obesity therapeutics, the specific mechanisms that switch on the appetite for sugar in states of satiety are unknown.

[1]Research Group Synaptic Transmission in Energy Homeostasis, Max Planck Institute for Metabolism Research, Cologne, Germany. [2]Policlinic for Endocrinology, Diabetes, and Preventive Medicine, University Hospital Cologne, Cologne, Germany. [3]Translational Neurocircuitry Group, Max Planck Institute for Metabolism Research, Cologne, Germany. [4]Global Drug Discovery, Novo Nordisk A/S, Måløv, Denmark. [5]Excellence Cluster on Cellular Stress Responses in Aging Associated Diseases (CECAD), University of Cologne, Cologne, Germany. [6]Institute of Zoology, Department of Biology, University of Cologne, Cologne, Germany. [7]Department of Brain Sciences, Weizmann Institute of Science, Rehovot, Israel. [8]Max Planck Florida Institute for Neuroscience, Jupiter, FL, USA. [9]INSERM U1114, Centre de Recherche en Biomédecine de Strasbourg, Université de Strasbourg, Strasbourg, France. [10]Global Drug Discovery, Novo Nordisk A/S, Lexington, MA, USA. [11]Faculty of Medicine, University of Cologne, Cologne, Germany.
*Corresponding author. Email: henning.fenselau@sf.mpg.de

Pro-opiomelanocortin (POMC) neurons of the arcuate nucleus (ARC) of the hypothalamus are a core element of the neurocircuitry that controls satiety. POMC neurons are activated in fed conditions and release alpha-melanocyte–stimulating hormone (α-MSH), a neuropeptide that binds to the melanocortin-4 receptor (MC4R) (*6*). Genetic deficiency of *Pomc* or *Mc4r* causes hyperphagia and massive obesity (*7*–*10*), whereas administration of α-MSH into the paraventricular hypothalamus (PVH) or activating second-order MC4R-expressing PVH neurons potently reduce feeding (*11*, *12*). These findings have led to the general consensus that POMC neurons promote satiety through excitatory α-MSH–MC4R signaling in the downstream PVH (*1*). However, POMC also serves as the precursor of β-endorphin, an opioid peptide whose action on the mu-opioid receptor (MOR) stimulates appetite (*13*, *14*). Moreover, blocking β-endorphin–MOR signaling with centrally acting opioid receptor antagonists decreases sugar intake, which is particularly prominent in ad libitum fed conditions (*15*–*17*). Because of the widespread projections of POMC neurons (*18*), we hypothesized that they might be the source of the endogenous opioid signal that actuates this sugar appetite in satiety states through a downstream brain region other than the PVH.

POMC neurons selectively inhibit paraventricular thalamic neurons

We first analyzed expression of the MOR in the projection sites of POMC neurons (Fig. 1A). We reasoned that high levels of MOR expression on postsynaptic neurons could predict POMC neuron–innervated regions with facilitated β-endorphin–MOR signaling. We performed fluorescence in situ hybridization (FISH) with a probe for the MOR gene (*Oprm1*) followed by immunohistochemistry (IHC) staining with a POMC antibody. MOR expression was detectable in all brain regions that contained POMC-expressing terminals (Fig. 1, A and B, and fig. S1, A and B). However, the paraventricular nucleus of the thalamus (PVT), a midline thalamic structure that has emerged as an important relay station of signals for feeding and motivated behavior (*19*–*22*), showed both the densest number of MOR-positive cells and the highest MOR expression intensity among all POMC-innervated sites (Fig. 1, A and B, and fig. S1, A and B). The high expression of MOR and POMC was found throughout the anterior to posterior extent of the PVT (fig. S1, C and D). Immunostainings in mice expressing an mCherry-tagged MOR (*23*) revealed the close proximity of β-endorphin–positive POMC neuron terminals and MORs in the PVT (fig. S2A).

To test how postsynaptic PVT neurons respond to POMC neuron terminal activation, we used an optogenetic-electrophysiology approach (Fig. 1C). Because neuropeptide release is favored during high-frequency firing (*24*–*26*), we virally expressed the kinetically fast optogenetic activator ChETA in POMC neurons (Fig. 1C and fig. S2B). High-frequency (20 Hz) photostimulation of ChETA-expressing POMC neuron terminals inhibited PVT neurons, as determined by a reduction in spontaneous firing rate and membrane hyperpolarization (Fig. 1, D and E, and fig. S2C). To assess the contribution of MOR signaling, we performed recordings in the presence of the opioid receptor antagonist naloxone or the selective MOR antagonist CTAP. Addition of either drug blocked the inhibitory action of POMC neuron terminal stimulation (Fig. 1E and fig. S2C), which demonstrates that MOR activation underlies the inhibitory action of the POMC→PVT circuit.

Because many POMC neurons express markers for glutamate and γ-aminobutyric acid (GABA) (*27*), we additionally investigated PVT neuron responses in the presence of blockers for these fast-acting neurotransmitters. Addition of synaptic blockers had no influence on PVT neuron inhibition upon POMC neuron terminal stimulation (Fig. 1E and fig. S2C). Consistent with the postsynaptic inhibition of PVT neurons upon MOR activation, bath application of the selective MOR agonist DAMGO hyperpolarized PVT neurons and decreased firing frequency (Fig. 1F and fig. S2D), without causing significant alterations in spontaneous glutamate and GABA release (fig. S2E).

We next measured the release of β-endorphin from POMC neuron terminals in the PVT. In ChETA-expressing mice, high-frequency optogenetic stimulation of POMC→PVT projections evoked β-endorphin secretion as assessed by enzyme-linked immunosorbent assay (ELISA) (fig. S2F). To examine the temporal features of β-endorphin–MOR signaling in the PVT, we used an optogenetic-imaging approach (Fig. 1G).

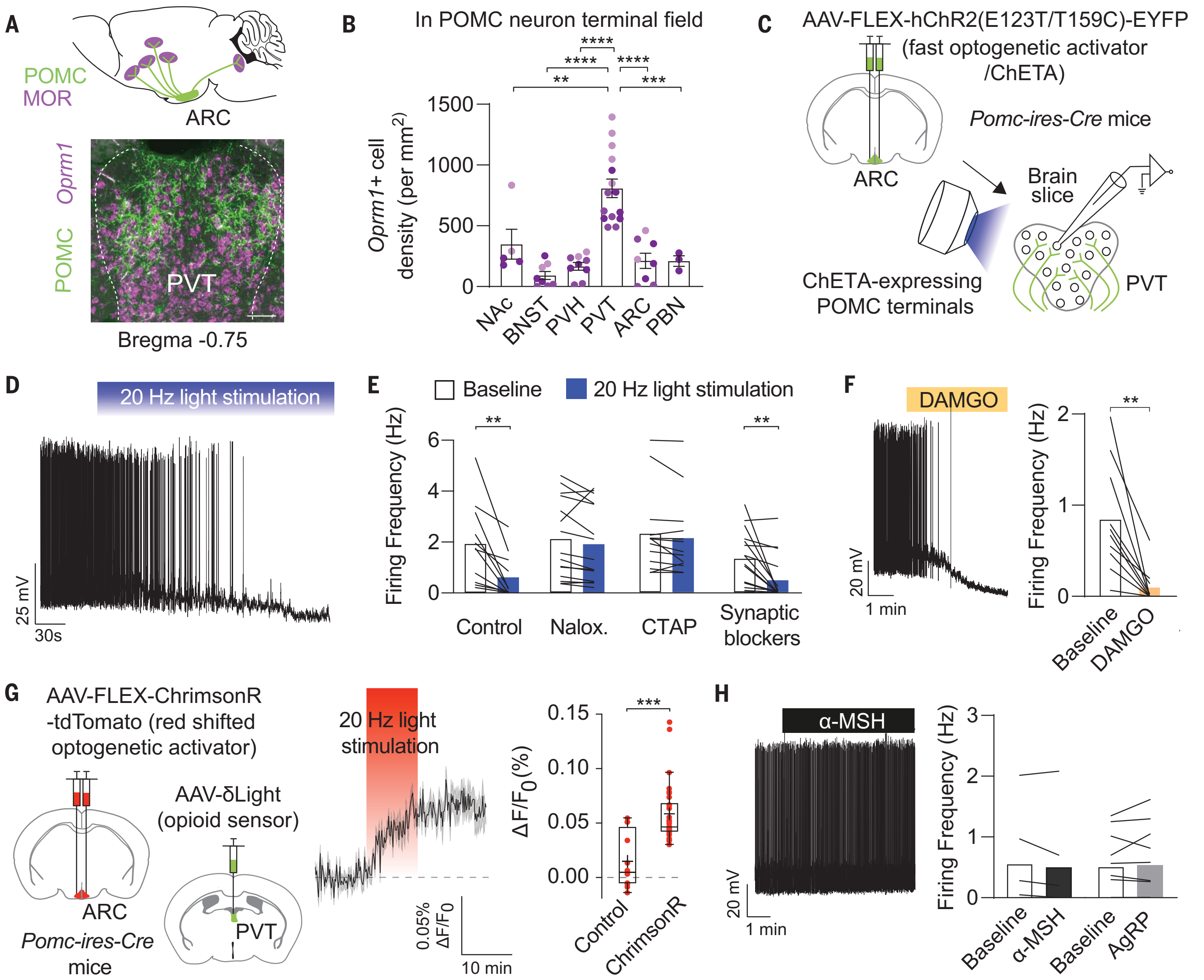

Fig. 1. POMC neurons silence PVT neurons through opioid signaling. (**A**) (Top) Schematic of the histological approach to characterize POMC neuron projection sites. (Bottom) Fluorescence image showing dense MOR expression (*Oprm1*) in the field of POMC neuron terminals in the PVT. Scale bar, 50 μm. (**B**) Analysis of MOR-expressing (*Oprm1*+) cells in POMC-innervated brain regions. BNST, bed nucleus of the stria terminalis; PBN, parabrachial nucleus. (**C**) (Top) A Cre-dependent viral approach to express the kinetically fast optogenetic activator ChETA for selective activation of POMC neuron terminals in the PVT. (Bottom) Patch clamp recordings were performed from PVT neurons in the field of ChETA-expressing terminals (EYFP+). EYFP, enhanced yellow fluorescent protein. (**D**) Current-clamp recordings from PVT neurons. Blue light (473 nm)–evoked photostimulation (20 Hz) of POMC neuron terminals inhibited action-potential firing and induced hyperpolarization. (**E**) Summary of firing frequency shows that POMC neuron terminal stimulation silenced PVT neurons. Addition of naloxone (Nalox.) or CTAP (D-Phe-Cys-Tyr-D-Trp-Arg-Thr-Pen-Thr-NH2) blocked the light-evoked inhibition of PVT neurons, whereas antagonists of glutamate and GABA-A receptors (CNQX, D-AP5, and bicuculline; synaptic blockers) had no effects. (**F**) Trace and summary of DAMGO-induced silencing of PVT neurons. DAMGO is a selective MOR agonist. (**G**) (Left) Viral approach to express the red-shifted, kinetically fast optogenetic activator ChrimsonR in POMC neurons and the opioid sensor δLight in the PVT. (Right) δLight responses to photostimulation of POMC neuron terminals in the PVT in the absence (control) and presence of ChrimsonR. (**H**) Trace and summary of firing frequency changes in PVT neurons in response to α-MSH or AgRP application. Data presented in the graphs in (B), (E), (F), and (H) represent means ± SEMs. Horizontal lines in box plots in (G) indicate medians, and crosses indicates means; whiskers were calculated using the Tukey method. Statistical analysis was performed using ordinary one-way analysis of variance (ANOVA), Tukey's multiple comparisons test (B), two-tailed paired Student's *t* test [(E), (F), and (H)], or Mann-Whiney *U* test (G). $^{**}P \leq 0.005$; $^{***}P \leq 0.001$; $^{****}P \leq 0.0001$. Number of mice for experiments: $N = 3$ (B); $N = 4$ (E); $N = 4$ (F); $N = 3$ (control) or 5 (ChrimsonR) (G); $N = 3$ (α-MSH) or 7 (AgRP) (H). (B) Data from each mouse are represented in the same shading (light purple for mouse 1, purple for mouse 2, and dark purple for mouse 3).

Stimulation of POMC neuron terminals in the PVT using the red light–shifted, kinetically fast optogenetic activator ChrimsonR produced a spatially restricted increase in fluorescence of the opioid sensor δLight (*28*) in a frequency-dependent manner (Fig. 1G and fig. S2, G to J).

We then considered the possibility that concomitant release of melanocortin neuropeptides could also be involved in transmission across the POMC→PVT synapse. We first analyzed the distribution of the MC4R, whose activation by POMC neuron–derived α-MSH activates postsynaptic neurons in the PVH (*11*, *29*, *30*). The PVT, unlike other POMC neuron projection sites, showed a near-complete absence of MC4Rs (fig. S3, A and B). In support of our histological findings, bath applications of α-MSH, or the endogenous MC4R antagonist agouti-related protein (AgRP), did not acutely alter the membrane potential or firing frequency of PVT neurons (Fig. 1H and fig. S3, C and D).

POMC neuron activity stimulates sated sugar intake

We next investigated the role of the POMC→PVT opioid circuit in feeding behavior regulation. In humans, surplus intake of high sugar–containing foods is particularly prominent as dessert. We therefore designed a behavioral paradigm that captures this phenomenon in mice (Fig. 2A). After an overnight fast, mice were provided access with chow diet for a refeeding period of 90 min, after which they were given access to either chow or a high sugar–containing diet (HSD) for 30 min. When chow was given during this 30-min dessert period, mice consumed an expected small amount, confirming the enhanced satiety state induced by the large amount of chow that they consumed during the 90-min refeeding period (Fig. 2B). By contrast, access to HSD during the dessert period potently stimulated feeding, increasing the caloric intake by more than sixfold (Fig. 2B). This vigorous stimulation of consumption of the high sugar–containing food was consistent across all mice.

To investigate the activity dynamics of the POMC→PVT circuit during the refeeding-dessert paradigm, we used fiber photometry and recorded GCaMP6s dynamics in POMC neuron terminals as a proxy for presynaptic activity (Fig. 2C and fig. S3E). During the initial 90-min refeeding period, presynaptic activity was only modestly increased, albeit the mice voraciously consumed chow (Fig. 2, D to F, and fig. S3F). By contrast, HSD consumption during the dessert period was accompanied by a strong activation of POMC→PVT neuron terminals (Fig. 2, D to F, and fig. S3F). The increase in presynaptic activity occurred rapidly upon HSD presentation and remained elevated during consumption (Fig. 2, D and E, and fig. S3F). Video analysis of head orientation revealed that presynaptic activity increased before the start of HSD consumption and aligned with the behavioral orientation toward the food pellet (Fig. 2F and fig. S3G). This suggests that activation of the POMC→PVT opioid circuit may be necessary for driving the appetitive and consummatory aspects of high sugar–containing food.

POMC neurons are strongly activated upon chow diet presentation and consumption in fasted mice—although at the level of their cell bodies in the ARC (*31*, *32*). To reexamine this, we recorded GCaMP6s dynamics in POMC neurons with optical fibers implanted above the ARC (fig. S3H). Consistent with previous observations (*31*, *32*), refeeding fasted mice

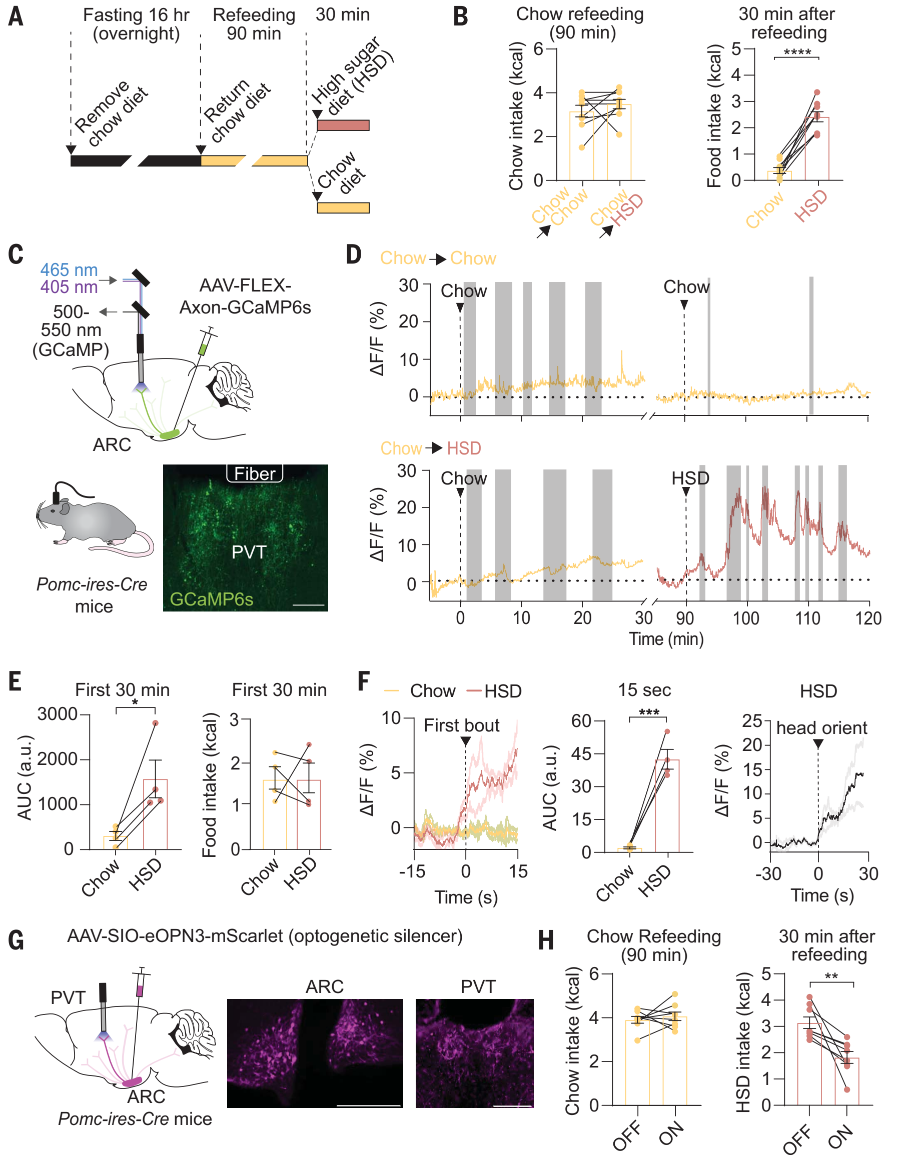

Fig. 2. POMC neuron activity drives sugar consumption after refeeding. (**A**) Schematic of experimental design for the refeeding-dessert paradigm to assess HSD intake in acutely sated mice. (**B**) (Left) Chow diet consumption (in calories) during the 90-min refeeding period was not altered when mice received either chow diet (Chow→Chow) or the HSD (Chow→HSD) during the 30-min dessert period. (Right) Caloric intake was significantly higher when mice received HSD instead of chow diet during the dessert period (30 min after refeeding). (**C**) (Top) A Cre-dependent viral approach to express the axon-targeted GCaMP6s construct in POMC neurons. The photometry fiber was implanted above the PVT for measuring GCaMP6s signals in POMC neuron terminals. (Bottom) Fluorescence image showing GCaMP6s expression in the PVT. Scale bar, 50 μm. (**D**) GCaMP6s fluorescence traces. (Top) A mouse received chow during both the refeeding period and the dessert period. (Bottom) A mouse received a chow during the refeeding period and HSD during the dessert period. Gray areas indicate eating bouts. (**E**) (Left) Quantification of fiber photometry responses during the first 30 min of chow diet consumption in the refeeding period and during the 30-min dessert period when mice consumed the HSD. AUC, area under the curve; a.u., arbitrary units. (Right) Chow diet intake during the first 30 min of the refeeding period and HSD intake during the first 30 min of the dessert period, assessed during the fiber photometry recordings. (**F**) (Left) POMC→PVT terminal fiber photometry responses aligned to time of the first feeding bout of chow diet and HSD. (Middle) Quantification of responses 15 s after alignment to the first feeding bout. (Right) Photometry responses aligned to the head orientation movement. (**G**) Cre-dependent viral approach to express the optogenetic inhibitor eOPN3 in POMC neurons. The optic fiber was implanted above the PVT. Fluorescence image shows eOPN3 (mScarlet) expression in the ARC and PVT. Scale bars, 100 μm (ARC) and 50 μm (PVT). (**H**) eOPN3-mediated inhibition of POMC→PVT projections reduced HSD intake during the dessert period after refeeding ("off," no light; "on," light stimulation). All data presented in the graphs represent means ± SEMs. Statistical analysis was performed using two-tailed paired Student's *t* test [(B), (E), (F), (H)]; *$P \leq 0.05$; **$P \leq 0.005$; ***$P \leq 0.001$; ****$P \leq 0.0001$.

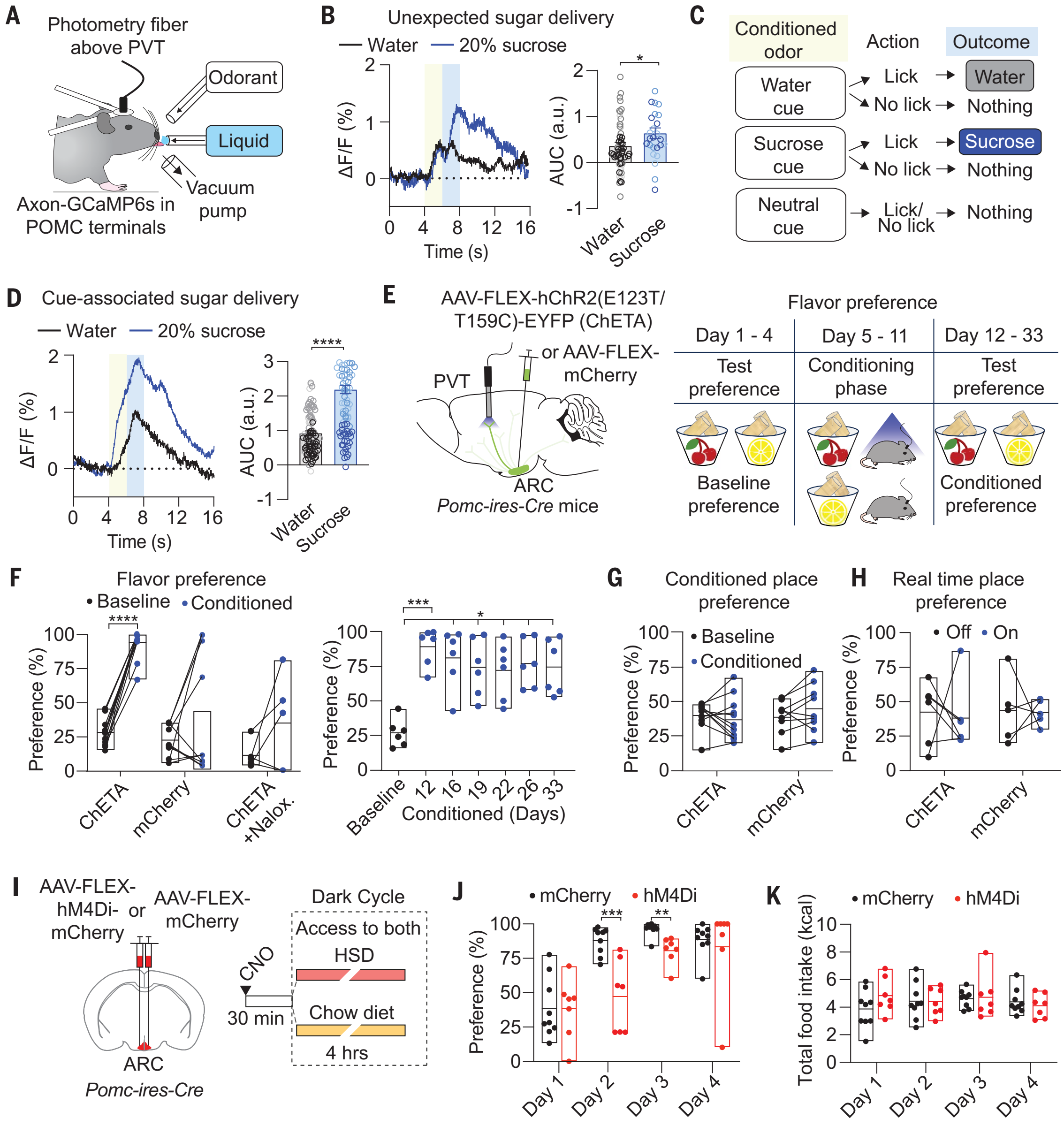

Fig. 3. The POMC→PVT circuit promotes the formation of sugar preference. (**A**) Schematic of the setup for in vivo measurement of fiber photometry responses in POMC→PVT terminals in head-fixed mice. The presented olfactory cues were removed by a vacuum pump. Licking was detected by a lickometer. (**B**) (Left) GCaMP6s fluorescence traces of water and sugar solution delivery within the same animal when completely naïve to the sugar solution. Yellow area indicates 2-s cue presentation window. Blue area indicates 2-s response window. (Right) Quantification of responses during olfactory cue presentation and delivery of solution (between 4 and 8 s). (**C**) Design of the go/no-go olfactory discrimination task for measuring responses to learned sugar-predicting cues. The cues were presented for 2 s followed by a 2-s response window. If mice responded with licking after offset of the water or sucrose predictive olfactory cue, they received 4 μl of water or the sugar solution (20% sucrose), respectively; they received nothing after the neutral olfactory cue. (**D**) (Left) Traces of water and sugar solution delivery within the same animal after training. (Right) Quantification of responses during cue presentation and delivery of solution (between 4 and 8 s) in trained mice. Colored areas indicate the 2-s presentation and 2-s response windows. (**E**) Experimental design for optogenetic stimulation of the POMC→PVT circuit (left). ChETA or mCherry was expressed in POMC neurons, and optical fibers were implanted above the PVT. Experimental design of the CFP assay is shown (right). (**F**) ChETA-expressing mice exhibited a significant preference for the flavored diet that was paired with blue light during the conditioning period, as compared with mCherry-expressing mice (day 12, first day of assessment; left). Naloxone administration before light illumination during the conditioning phase inhibited the diet preference change in ChETA-expressing mice. Preference changes for the flavored diet that was paired with light lasted until the end of the assessment (day 33). (**G**) Changes in preference for the chamber that was paired with light illumination during the conditioning period in mice expressing ChETA or mCherry in POMC neurons. (**H**) Place preference changes during a real-time place preference assay. (**I**) Experimental design for chemogenetic inhibition of POMC neurons. Mice expressing hM4Di or mCherry (control) were injected with CNO before being given access to HSD and chow diet during the first 4 hours of the dark cycle. (**J**) hM4Di-expressing mice exhibited a time delay in preference for consuming the HSD, as compared with mCherry-expressing mice (all mice received CNO). (**K**) hM4Di- and mCherry-expressing mice consumed the same total amount of food (in calories). All data presented in the graphs represent means ± SEMs. Statistical analysis was performed using two-tailed paired Student's t test [(B), (D), (G), (H), (J), and (K)], two-way ANOVA with Sidak's multiple comparisons test [(F), left], or ordinary one-way ANOVA followed by Tukey's multiple comparisons test [(F), right]; $^{*}P \leq 0.05$; $^{**}P \leq 0.01$; $^{****}P \leq 0.0001$. Number of mice for fiber photometry experiments: $N = 3$. Number of trials: water (72), sucrose (26) (B); water (98), sucrose (97) (D).

with chow diet potently increased POMC neuron activity in the ARC (fig. S3I).

The activation of POMC→PVT projections during HSD consumption and subsequent postsynaptic opioid signaling should inhibit PVT neurons, which has been associated with driving food consumption (*19*, *20*, *22*). To explore this, we optogenetically inhibited POMC neuron terminals in the PVT (Fig. 2G and fig. S4A). Photoinhibition of POMC→PVT projections in fasted, eOPN3-expressing mice reduced HSD intake during the dessert period without altering chow diet intake during the 90-min refeeding period (Fig. 2H and fig. S4B). Chemogenetic inhibition of POMC neurons expressing hM4Di using clozapine *N*-oxide (CNO) similarly reduced HSD intake (fig. S4, C to E). In contrast to this acute sugar appetite–suppressing effect, 48-hour intake of chow diet was increased by continuous chemogenetic inhibition of POMC neurons (fig. S4F). This finding is consistent with the previously reported, delayed anorexigenic function of POMC neurons and the melanocortin neuropeptides that they release (*29*, *33*, *34*).

Sugar cues rapidly activate POMC→PVT projections

We next sought to precisely determine the relationship between activation of the POMC→PVT circuit and sugar consumption. Because mice were acclimated to the HSD for our refeeding-dessert paradigm (Fig. 2A) before the experiments to avoid food neophobia, we considered the possibility that the immediate increase in presynaptic activity upon HSD detection is preingestive and that its rapid activation involves a learning process. We therefore recorded presynaptic activity in head-fixed animals, which were trained to perform a go/no-go discrimination task (Fig. 3A). We first investigated how delivery of a sugar solution to an olfactory cue that predicts water access influences presynaptic POMC→PVT terminal activity. Water-deprived mice were presented with one of two distinct olfactory cues, after which licking to the correct (water) cue led to delivery of a water reward (4-μl drop) (fig. S5, A and B). In well-trained mice that selectively licked to the water cue (fig. S5, A and B), unexpected delivery of a high sugar–containing solution (20% sucrose) resulted in a rapid increase in POMC→PVT terminal activity (Fig. 3B). This response emerged during the onset of solution delivery (Fig. 3B), which indicates that signals from the oral cavity rapidly reach POMC neurons that project to the PVT.

We also observed a small, yet consistent increase in presynaptic activity to the water cue, which emerged shortly after stimulus onset (Fig. 3B). This unexpected rapid activation suggests that POMC neurons, which regulate various metabolic processes in addition to feeding (*35*), also integrate water-related signals, at least in a state of mild dehydration. Furthermore, the increased activity preceding water access indicates that the thirsty mice learned to associate the otherwise arbitrary (olfactory) cue with water availability.

We next recorded presynaptic activity in response to cues predicting sugar access by including an additional olfactory cue after which licking led to the delivery of the high sugar–containing solution (Fig. 3C). In trained mice, activity of POMC→PVT terminals profoundly increased to sugar cue presentation and solution consumption (Fig. 3D). This increase in activity immediately emerged during presentation of the cue and consistently exceeded the response to the water cue (Fig. 3D). This difference was not because of differential

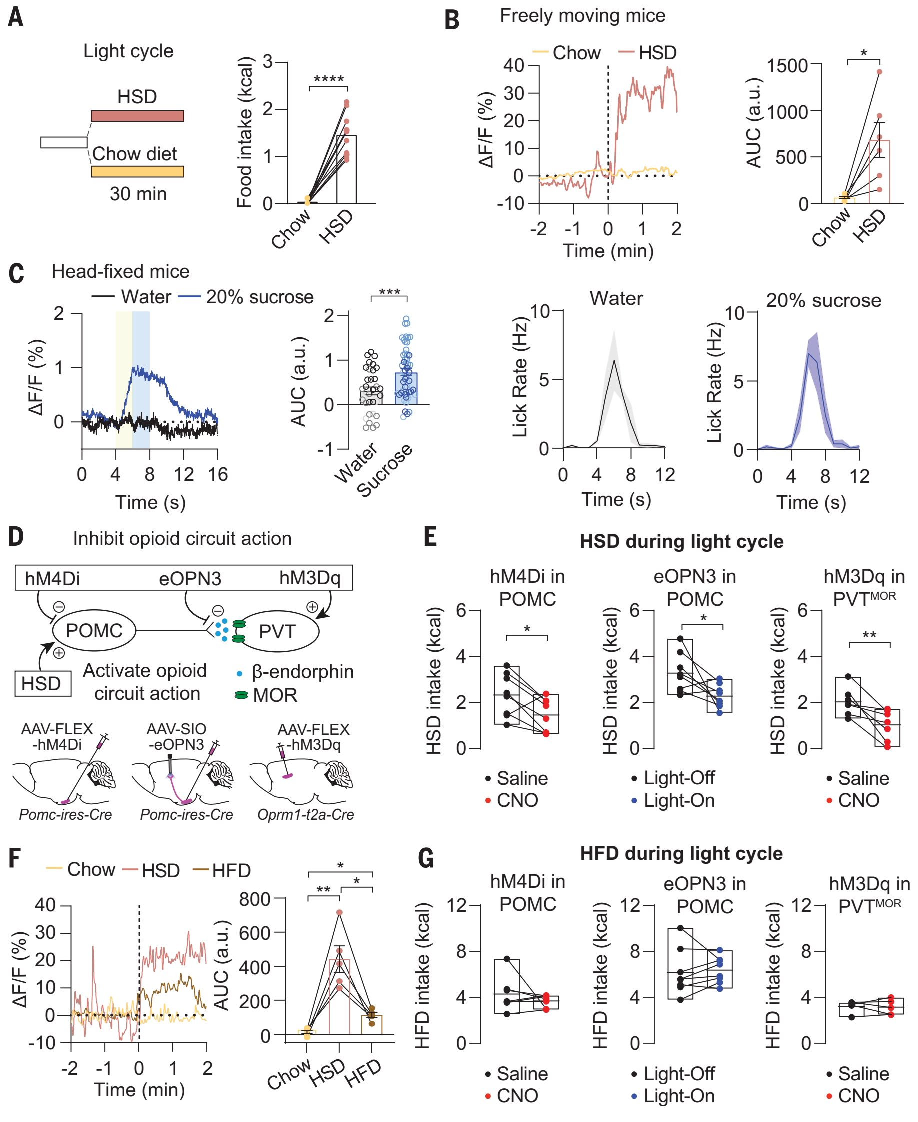

Fig. 4. POMC→PVT circuit activity drives sated sugar—but not fat—consumption. (**A**) Providing HSD at the onset of the light cycle resulted in voracious feeding in mice. (**B**) POMC→PVT terminals are profoundly activated upon HSD presentation and consumption in freely behaving mice. (Left) Fiber photometry recordings from POMC→PVT terminals expressing GCaMP6s from the same mouse. (Right) Mean responses to chow diet and HSD presentation demonstrating increased presynaptic activity within the first 2 min after food presentation. (**C**) In fed and hydrated mice, presentation of sugar cues and sugar solution consumption evoked rapid responses in POMC→PVT neuron terminals in head-fixed mice. (Left) Traces of water and sugar solution delivery within the same animal. (Middle) Quantification of fiber photometry responses upon delivery of water and the sugar solution between 4 and 8 s. (Right) Licking behavior during the response window using 1-s bins. Mice licked similarly to both sugar and water cues. (**D**) Schematic of chemo- and optogenetic manipulations of the POMC→PVT circuit. (**E**) HSD intake in fed mice is significantly reduced by hM4Di-mediated inhibition of POMC neurons, eOPN3-mediated optogenetic inhibition of POMC→PVT projections, and hM3Dq-mediated activation of PVT^{MOR} neurons. (**F**) POMC→PVT terminals are activated upon HFD presentation and consumption in fed mice, but to a much lesser extent than for HSD. (Left) Fiber photometry recordings from POMC→PVT terminals expressing GCaMP6s from the same mouse. (Right) Mean responses to chow diet, HSD, and HFD within the first 2 min after food presentation. (**G**) Effects of chemo- and optogenetic manipulations of the POMC→PVT circuit on HFD consumption in fed mice. All data presented in the graphs represent means ± SEMs. Statistical analysis was performed using two-tailed paired Student's *t* tests [(A) to (C), (E), and (G)] or one-way ANOVA followed by Tukey's multiple comparisons test (F). Number of head-fixed mice for fiber photometry experiments: $N = 3$. Number of trials: water (50), sucrose (49). *$P \le 0.05$; **$P \le 0.01$; ***$P \le 0.001$; ****$P \le 0.0001$.

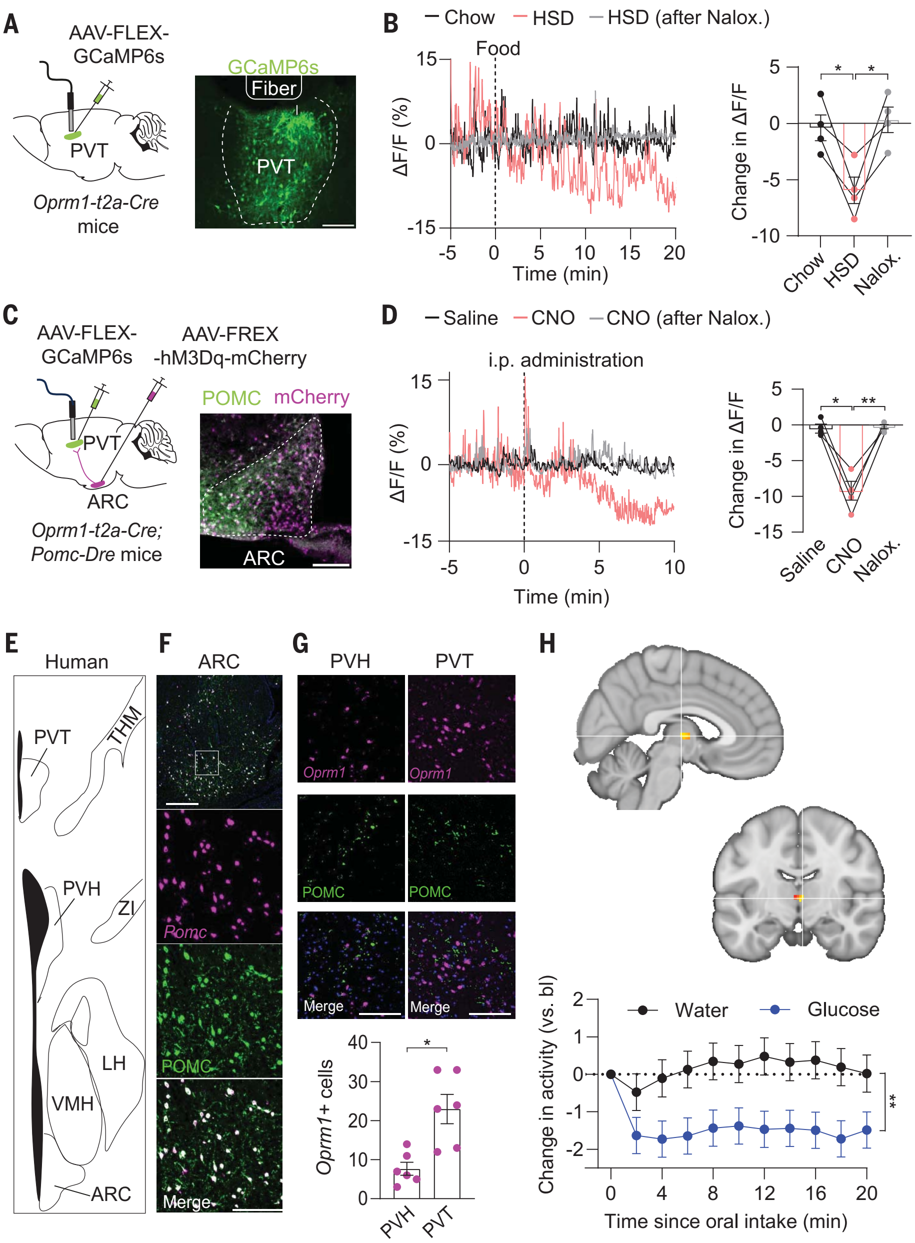

Fig. 5. Sugar-evoked responses in the PVT in mice and humans. (**A**) AAV-FLEX-GCaMP6s was injected into the PVT of *Oprm1-t2a-Cre* mice to determine real-time activity changes of PVT^{MOR} neurons using fiber photometry. Fluorescence image shows GCaMP6s expression and photometry fiber placement in the PVT. Scale bar, 50 μm. (**B**) (Left) GCaMP6s fluorescence traces in a single mouse upon delivery of chow or HSD recorded during the onset of the light cycle, and upon delivery of HSD with prior naloxone administration. (Right) Comparison of PVT^{MOR} neuron responses 20 min after food presentation. (**C**) AAV-FLEX-GCaMP6s was injected into the PVT and AAV-FREX-hM3Dq-mCherry was injected into the ARC of *Oprm1-t2a-Cre; Pomc-Dre* mice to activate POMC neurons while measuring PVT^{MOR} neuron activity. Fluorescence image shows hM3Dq (mCherry) and POMC expression in the ARC. Scale bar, 50 μm. (**D**) GCaMP6s fluorescence traces in a single hM3Dq-expressing mouse upon injection of saline or CNO and upon injection of CNO after naloxone injection. i.p., intraperitoneal. (Right) Comparison of PVT^{MOR} neuron responses 10 min after injection of saline or CNO. (**E**) Diagram derived from the human brain atlas. THM, thalamus; ZI, zona incerta; LH, lateral hypothalamus; VMH, ventromedial hypothalamic nucleus. (**F**) Fluorescence images showing POMC protein and *Pomc* mRNA expression in the ARC. (**G**) (Top) POMC protein (terminals) and *Oprm1* mRNA expression in the PVH and PVT. (Bottom) Quantification of *Oprm1*+ cells in POMC-innervated areas in human brain samples is shown. (**H**) Change in human PVT activity after oral intake of water (black) or glucose solution (blue), relative to the baseline (bl) before the intake. The consumption of a glucose solution relative to water reduced the BOLD signal in the PVT across 10 consecutive 2-min time intervals (up to 20 min after oral intake; $P < 0.05$, FWE-corrected at the voxel level within the PVT mask; bars show means and standard errors; activity maps overlayed on MNI152 brain template). All data presented in the graphs represent means ± SEMs. Statistical analysis was performed using ordinary one-way ANOVA followed by Tukey's multiple comparisons test [(B) and (D)] or two-tailed Student's *t* test (G) and t-contrasts followed by an omnibus test (H); $^{*}P \le 0.05$; $^{**}P \le 0.01$. Scale bars, 300 μm (F) and 50 μm (G). (G) Number of human donors: $N = 2$; number of brain slices per brain area: $n = 6$. (H) Number of human subjects, $N = 30$.

licking behavior (fig. S5C). Thus, although oral sugar sensing is initially sufficient to increase presynaptic activity, sugar cues and associated consumption expedite and strongly potentiate the synaptic response. The activation of POMC→PVT terminals scaled with the amount of sucrose in the solution, and even a 0.7% sucrose solution evoked an increase in presynaptic activity similar to that evoked by the 20% sucrose solution (fig. S5, D to F).

The POMC→PVT circuit promotes the formation of sugar preference

The learning-induced increase in presynaptic activity suggests that activation of the POMC→PVT opioid circuit may contribute to creating a sugar preference and inducing sugar consumption. We used a conditioned flavor preference (CFP) test combined with optogenetic manipulation (Fig. 3E). Mice expressing ChETA in POMC neurons and control mice expressing mCherry were habituated to lemon- and cherry-flavored chow diets, which were isocaloric, had no added sugar, and were identical in their nutrient composition. After determining the initial preference, POMC→PVT terminals were photostimulated during consumption of the diet with the initially less-preferred flavor (Fig. 3E). This approach, where circuit activation is paired with a flavored chow diet, circumvents potential confounding effects of sweet taste or postingestive sugar sensing on preference formation (*36–38*). ChETA-expressing mice showed a robust increase in preference for the photostimulation-paired diet after conditioning (Fig. 3F and fig. S6, A and B). This preference formation required opioid signaling because naloxone administration before the conditioning sessions prevented the behavioral change induced by photostimulation in ChETA-expressing mice (Fig. 3F). Measurement of diet preference over the following weeks demonstrated that the behavioral change persisted until the end of assessment (Fig. 3F and fig. S6C).

We additionally investigated the possibility that activation of the POMC→PVT opioid circuit is sufficient in eliciting a general preference behavior analogous to the activation of midbrain dopamine neurons. We first used the conditioned place preference paradigm to test the behavioral conditioning effects of photostimulation on environmental cues. After habituation and determining the initial preference, optogenetic stimulation of POMC→PVT terminals was paired with the initially less-preferred chamber of the apparatus (fig. S6D). ChETA-expressing mice did not develop a place preference after conditioning (Fig. 3G). To investigate whether the activation of POMC→PVT terminals is, by itself, sufficient to induce a positive valence, we performed a real-time place preference test. ChETA-expressing mice did not spend more time in the light-stimulated compartment (Fig. 3H), nor did we observe alterations in locomotor activity (fig. S6, E and F). Together, these data demonstrate that the POMC→PVT circuit is selectively important for entraining dietary preferences.

If activation of the POMC→PVT opioid circuit upon sugar detection is involved in creating a preference for high sugar–containing foods, then silencing it should diminish the formation of sugar preference. Mice were given a choice between HSD and chow diet at the onset of the dark cycle while chemogenetically inhibiting POMC neurons (Fig. 3I). CNO-treated control mice (mCherry-expressing) developed a strong preference for HSD already on the second day (Fig. 3J). By contrast, CNO- and hM4Di-mediated inhibition of POMC neurons delayed preference formation (Fig. 3J). Chemogenetic inhibition of POMC neurons did not alter total food consumption during the 4-hour experimental period (Fig. 3K). Consistent with the endogenous opioid action of the POMC→PVT circuit, administration of naloxone also delayed sugar preference formation without affecting total caloric consumption (fig. S6, G to I).

The POMC→PVT circuit drives sugar—but not fat—intake beyond caloric needs

High sugar–containing foods are not only favorably consumed after meals, when satiety is triggered through short-term feedback signals from the gut, but also in states of caloric surplus, when long-term satiety signals (such as leptin) are critical factors that determine appetite control (*1*). To test whether activity of the POMC→PVT opioid circuit also contributes to the regulation of sugar consumption in states of caloric surplus, we provided mice access to HSD at the beginning of the light cycle (Fig. 4A)—a time when mice are in a physiologically fed state from the food that they consumed during the dark cycle, accompanied by elevated levels of leptin (fig. S7A). Access to HSD for 30 min in acclimated mice induced voracious feeding (Fig. 4A and fig. S7B). Recording GCaMP6 signals in POMC→PVT neuron terminals revealed that HSD presentation and subsequent consumption resulted in a profound increase in presynaptic activity, which emerged rapidly after pellet delivery (Fig. 4B). Consistent with being in a state of caloric surplus, mice did not consume chow (Fig. 4A), and chow presentation did not elicit alterations in presynaptic activity (Fig. 4B and fig. S7C). In addition to these experiments in freely behaving mice, we also measured presynaptic activity in head-fixed animals. In fed and fully hydrated mice (fig. S7D), cue presentation and subsequent consumption of the high-sugar solution induced time-locked increases in presynaptic activity (Fig. 4C). The responses to the water cues that we observed in dehydrated mice were strongly diminished (Fig. 3, B and D, and Fig. 4C), which indicates that they specifically relate to the hydration state.

We next tested whether activity of the POMC→PVT opioid circuit is necessary for driving sugar consumption in states of caloric surplus (Fig. 4D). Chemogenetically inhibiting POMC neurons and optogenetic silencing of POMC→PVT projections diminished HSD intake at the onset of the light cycle (Fig. 4, D and E, and fig. S7, E to H). Selective chemogenetic stimulation of MOR-expressing neurons in the PVT (PVT^{MOR} neurons), which counteracts the inhibitory opioid action of the POMC→PVT circuit, also diminished HSD intake in fed mice (Fig. 4, D and E, and fig. S7, E to H).

Because high fat–containing foods are likewise consumed beyond caloric needs, we assessed the contribution of POMC→PVT circuit activity in regulating intake of a high-fat diet (HFD). Access to HFD for 30 min at the beginning of the light cycle potently stimulated feeding, with mice consuming more calories than when provided HSD (fig. S7I). Recording GCaMP6s dynamics in POMC→PVT terminals showed that HFD presentation and consumption evoked an increase in presynaptic activity; yet, this response was significantly smaller in magnitude than the response evoked by HSD presentation and consumption (Fig. 4F). In fed mice, HFD intake was not altered by chemogenetic inhibition of POMC neurons, optogenetic silencing of POMC→PVT projections, or activation of PVT^{MOR} neurons (Fig. 4G and fig. S7J). Inhibition of the POMC→PVT circuit also failed to reduce HFD intake when provided for 30 min after refeeding during the dessert period (fig. S7, K and L). We then determined the possibility that activity of the POMC→PVT opioid circuit contributes to the regulation of feeding at the onset of the dark cycle when mice are naturally energy depleted because of their circadian feeding rhythm. Chemogenetically inhibiting POMC neurons, activating PVT^{MOR} neurons, and blocking endogenous opioid signaling with naloxone all failed to acutely (within 3 hours) alter chow diet intake (fig. S7, M and N). Inhibition of the circuit also did not alter HSD intake at the onset of the dark cycle (fig. S7O). Thus, activity of the POMC→PVT opioid circuit is of particular importance in driving the specific appetite for sugar in states of satiety.

Dysregulation of the PVT by exogenous opioids mediates opioid withdrawal symptoms, and PVT neurons invigorate sugar seeking during unexpected reward omission (*39*–*41*). We therefore sought to precisely determine the relationship between endogenous opioid signaling in PVT neurons and sugar consumption. We first used fiber photometry to record the activity of PVT^{MOR} neurons expressing GCaMP6s (Fig. 5A). Consistent with the activation of the inhibitory POMC→PVT opioid circuit, presentation and subsequent consumption of HSD—but not chow diet—inhibited PVT^{MOR} neurons in fed and acutely sated mice (Fig. 5B and fig. S8, A and B), an effect that was blocked by preadministration of naloxone (Fig. 5B and fig. S8, C and D). To directly test whether POMC neurons underlie this endogenous opioid response, we recorded PVT^{MOR} neurons while selectively manipulating POMC neuron activity (Fig. 5C). We injected Cre-dependent GCaMP6s adeno-associated virus (AAV) into the PVT and Dre-dependent hM3Dq AAV into the ARC of *Oprm1-t2a-Cre; Pomc-Dre* mice and placed fiberoptic implants above the PVT (Fig. 5C). CNO- and hM3Dq-induced activation of POMC neurons significantly decreased PVT neuron activity (Fig. 5D and fig. S8E). This inhibition was abolished by naloxone pretreatment (Fig. 5D and fig. S8E). If activation of the POMC→PVT opioid circuit underlies the inhibition of PVT^{MOR} neurons, then activation of POMC neurons preceding HSD access should occlude the response. CNO administration in doubly transduced, hM3Dq-expressing mice significantly attenuated the HSD-evoked response in PVT^{MOR} neurons (fig. S8F). Because exogenous opioid action maintains seeking behavior via PVT neurons that project to the nucleus accumbens (NAc) (*39*, *41*, *42*), we used *Oprm1-t2a-cre; D1r-tdTomato* mice to assess synaptic connection characteristics between PVT^{MOR} and NAc medium spiny neurons (MSNs) expressing dopamine receptor D1 (tdTomato+) and neurons putatively expressing the D2 receptor (tdTomato–) in the NAc shell. Electrophysiological recordings revealed that PVT^{MOR} neurons provide strong excitatory input to both D1R+ and putative D2R NAc neurons with similar synaptic characteristics (fig. S9, A to G).

Sugar consumption decreases human PVT activity

We next determined whether the POMC→PVT opioid circuit is conserved in humans. To histologically assess the proximity of POMC neuron terminals and PVT^{MOR} neurons, we costained brain sections from human tissues for POMC

(IHC) and *Oprm1* (FISH). We confirmed faithful and efficient binding of the POMC antibody to POMC neurons in the human ARC as determined by FISH (Fig. 5, E and F). In the PVT, POMC-expressing terminals were directly adjacent to a large fraction of MOR-expressing cell bodies; by contrast, density of MOR-expressing cells in the terminal field was lower in the PVH (Fig. 5G). Finally, to determine activity changes in the human PVT during sugar consumption, we performed functional magnetic resonance imaging with normal-weight participants (Fig. 5H). Consumption of a high sugar–containing solution reduced the subsequent blood oxygen level–dependent (BOLD) response in the PVT relative to the baseline before consumption (Fig. 5H) (one-sampled t test for the contrast –glucose: t = 3.47; family-wise error corrected for multiple comparisons within the PVT mask: $P_{\text{FWE-corr}}$ = 0.006; cluster size = 37; peak at x = 4, y = −12, z = 4), even in a direct comparison with water intake (paired t test for the contrast water > glucose: t = 3.12; $P_{\text{FWE-corr}}$ = 0.016; cluster size = 13; peak at x = 4, y = −12, z = 6). The opposite contrasts (–water and glucose > water) did not reveal suprathreshold clusters, even at a more lenient statistical threshold without correction for multiple comparisons.

Discussion

POMC neurons of the hypothalamus produce numerous neuropeptides, including α-MSH and β-MSH, which exert well-established satiety-promoting and obesity-preventing actions because of their binding to the MC4R (*6*, *43*). Additionally, POMC neurons are the sole brain source of the potent MOR agonist β-endorphin (*44*), which exhibits broad effects on reward-seeking and food-intake behavior when administered into different brain regions (*14*, *45–48*). Thus, it is important to establish where in the brain POMC neurons use these functionally different melanocortin and opioid neuropeptides as endogenous neuronal messenger. In this work, we found that the PVT constitutes a projection area of POMC neurons, where their β-endorphin–MOR signaling selectively silenced postsynaptic neurons. The POMC→PVT circuit was rapidly engaged by sugar detection in the oral cavity and remained activated during sugar consumption. POMC→PVT activity increased upon cues instructive for sugar acquisition, which involved the learned association of environmental information with consumption. We probed the function of this rapid activity regulation, which was particularly pronounced in fed conditions. Our experiments unexpectedly revealed that POMC neurons and downstream PVT^{MOR} neurons are required to promote surplus consumption of high sugar–containing food.

Together, our data support a model where POMC neuron activation in the fed state actuates two, nearly opposing processes, namely (i) the known decrease in total food intake by activating satiety-promoting PVH neurons through MC4R signaling (*11*, *12*, *49*) and (ii) the herein described stimulation of sugar intake by inhibiting PVT neurons through MOR signaling. The orexigenic action of POMC neurons is specific to states of caloric sufficiency or surplus (Figs. 2 and 4) (*13*, *50*) and may thus act complementary to endogenous opioids from brainstem enkephalin neurons, which also promote consummatory behavior but in states of caloric deprivation (*50*). It would therefore be of interest to define how the state-dependent activity of these endogenous opioid systems interacts with dopamine signaling in the mesoaccumbens circuitry, which serves to control reward-driven behavior and association learning (*3*, *51*). Notably, MC4R-deficient mice and humans exhibit significant alterations in preferences for high sugar–containing foods (*52*, *53*), which raises the possibility that interactions between melanocortin and opioid signaling in response to POMC neuropeptide release further modulate macronutrient preferences.

The stimulation of sugar appetite by POMC neurons may have been evolutionary advantageous when food was scarce because sugars, when available, can be readily absorbed and metabolized for the synthesis of glycogen and lipids as energy reserve. However, it seems plausible that recurring activation of POMC→PVT opioid signaling when food and sugar are abundant could manifest compulsive-like eating behavior or even binge eating of sugar. The PVT has emerged as a key brain node underlying sugar seeking and feeding behavior (*20*, *21*, *40*), and its GABA-mediated inhibition evokes overeating and obesity development (*19*). Engagement of PVT neuron activity by exogenous opioids facilitates drug-seeking behavior and the expression of opioid withdrawal (*39*, *41*, *54*). Therefore, our findings may revive investigations on endogenous opioid signaling in sugar appetite regulation (*55*) and accelerate further investigations on the POMC→PVT circuit for the development of new obesity therapeutics.

REFERENCES AND NOTES

1. M. L. Andermann, B. B. Lowell, *Neuron* **95**, 757–778 (2017).
2. M. H. Tschöp, J. M. Friedman, *Sci. Transl. Med.* **15**, eadh4453 (2023).
3. I. E. de Araujo, M. Schatzker, D. M. Small, *Annu. Rev. Psychol.* **71**, 139–164 (2020).
4. E. C. O'Connor *et al.*, *Neuron* **88**, 553–564 (2015).
5. J. H. Jennings *et al.*, *Cell* **160**, 516–527 (2015).
6. M. J. Krashes, B. B. Lowell, A. S. Garfield, *Nat. Neurosci.* **19**, 206–219 (2016).
7. C. Vaisse, K. Clement, B. Guy-Grand, P. Froguel, *Nat. Genet.* **20**, 113–114 (1998).
8. G. S. Yeo *et al.*, *Nat. Genet.* **20**, 111–112 (1998).
9. L. Yaswen, N. Diehl, M. B. Brennan, U. Hochgeschwender, *Nat. Med.* **5**, 1066–1070 (1999).
10. D. Huszar *et al.*, *Cell* **88**, 131–141 (1997).
11. A. S. Garfield *et al.*, *Nat. Neurosci.* **18**, 863–871 (2015).
12. S. Q. Giraudo, C. J. Billington, A. S. Levine, *Brain Res.* **809**, 302–306 (1998).
13. M. Koch *et al.*, *Nature* **519**, 45–50 (2015).
14. L. D. McKay, N. J. Kenney, N. K. Edens, R. H. Williams, S. C. Woods, *Life Sci.* **29**, 1429–1434 (1981).
15. M. Apfelbaum, A. Mandenoff, *Pharmacol. Biochem. Behav.* **15**, 89–91 (1981).
16. J. Cleary, D. T. Weldon, E. O'Hare, C. Billington, A. S. Levine, *Psychopharmacology* **126**, 110–114 (1996).
17. A. S. Levine, D. T. Weldon, M. Grace, J. P. Cleary, C. J. Billington, *Am. J. Physiol.* **268**, R248–R252 (1995).
18. D. Wang *et al.*, *Front. Neuroanat.* **9**, 40 (2015).
19. X. Zhang, A. N. van den Pol, *Science* **356**, 853–859 (2017).
20. J. N. Betley, Z. F. H. Cao, K. D. Ritola, S. M. Sternson, *Cell* **155**, 1337–1350 (2013).
21. Y. Livneh *et al.*, *Nature* **546**, 611–616 (2017).
22. T. R. Stratford, D. Wirtshafter, *Brain Res.* **1490**, 128–133 (2013).
23. O. Gardon *et al.*, *Neuroscience* **277**, 595–609 (2014).
24. A. Dutton, R. E. J. Dyball, *J. Physiol.* **290**, 433–440 (1979).
25. C. Schöne, J. Apergis-Schoute, T. Sakurai, A. Adamantidis, D. Burdakov, *Cell Rep.* **7**, 697–704 (2014).
26. M. Cazalis, G. Dayanithi, J. J. Nordmann, *J. Physiol.* **369**, 45–60 (1985).
27. B. C. Jarvie, S. T. Hentges, *J. Comp. Neurol.* **520**, 3863–3876 (2012).
28. C. Dong *et al.*, *Nat. Neurosci.* **27**, 1844–1857 (2024).
29. H. Fenselau *et al.*, *Nat. Neurosci.* **20**, 42–51 (2017).
30. M. Ghamari-Langroudi, D. Srisai, R. D. Cone, *Proc. Natl. Acad. Sci. U.S.A.* **108**, 355–360 (2011).
31. Y. Chen, Y. C. Lin, T. W. Kuo, Z. A. Knight, *Cell* **160**, 829–841 (2015).
32. C. Brandt *et al.*, *Cell* **175**, 1321–1335.e20 (2018).
33. C. Zhan *et al.*, *J. Neurosci.* **33**, 3624–3632 (2013).
34. Y. Aponte, D. Atasoy, S. M. Sternson, *Nat. Neurosci.* **14**, 351–355 (2011).
35. J. C. Brüning, H. Fenselau, *Science* **381**, eabl7398 (2023).
36. H. E. Tan *et al.*, *Nature* **580**, 511–516 (2020).
37. I. E. de Araujo *et al.*, *Neuron* **57**, 930–941 (2008).
38. K. L. Buchanan *et al.*, *Nat. Neurosci.* **25**, 191–200 (2022).
39. Y. Zhu, C. F. R. Wienecke, G. Nachtrab, X. Chen, *Nature* **530**, 219–222 (2016).
40. F. H. Do-Monte, A. Minier-Toribio, K. Quiñones-Laracuente, E. M. Medina-Colón, G. J. Quirk, *Neuron* **94**, 388–400.e4 (2017).
41. K. M. Vollmer *et al.*, *Nat. Commun.* **13**, 6865 (2022).
42. J. M. Otis *et al.*, *Neuron* **103**, 423–431.e4 (2019).
43. Y. S. Lee *et al.*, *Cell Metab.* **3**, 135–140 (2006).
44. C. Toda, A. Santoro, J. D. Kim, S. Diano, *Annu. Rev. Physiol.* **79**, 209–236 (2017).
45. C. A. Baile, C. L. McLaughlin, M. A. Della-Fera, *Physiol. Rev.* **66**, 172–234 (1986).
46. B. A. Gosnell, J. E. Morley, A. S. Levine, *Brain Res.* **369**, 177–184 (1986).
47. M. Amalric, E. J. Cline, J. L. Martinez Jr., F. E. Bloom, G. F. Koob, *Psychopharmacology* **91**, 14–19 (1987).
48. V. P. Bakshi, A. E. Kelley, *J. Pharmacol. Exp. Ther.* **265**, 1253–1260 (1993).
49. N. Balthasar *et al.*, *Cell* **123**, 493–505 (2005).
50. D. C. Castro *et al.*, *Nature* **598**, 646–651 (2021).
51. S. J. Gershman, N. Uchida, *Nat. Rev. Neurosci.* **20**, 703–714 (2019).
52. A. A. van der Klaauw *et al.*, *Nat. Commun.* **7**, 13055 (2016).
53. B. L. Panaro, R. D. Cone, *Proc. Natl. Acad. Sci. U.S.A.* **110**, 7050–7055 (2013).
54. P. C. Keyes *et al.*, *Neuron* **107**, 1113–1123.e4 (2020).
55. P. K. Olszewski, A. S. Levine, *Physiol. Behav.* **91**, 506–512 (2007).
56. M. Minere *et al.*, Thalamic opioids from POMC satiety neurons switch on sugar appetite, version 1_28.08.2024, Zenodo (2024); https://doi.org/10.5281/zenodo.13385228.

ACKNOWLEDGMENTS

We thank M.-L. Dreisow and R. Beck for their excellent technical assistance. K. Schöfisch and R. Braun provided invaluable help for animal care. We appreciate the scientific counsel and input from P. N. Mirabella, S. Dodt, E. Ciglieri, and J. Brüning. We thank J. B. Johansen and J. Juul from Novo Nordisk A/S for technical support on human brain histology. We also acknowledge C. Smith at the Edinburgh Brain and Tissue Bank for providing human brain tissue samples. **Funding:** This study was supported by the European Research Council (ERC) under the European Union Horizon 2020 Research and Innovation Programme, grant agreement ID 851778 (H.F.); collaboration agreements with Novo Nordisk (Denmark) (H.F.); Deutsche Forschungsgemeinschaft (DFG, German Research Foundation), grants 409551513 (H.F.) and 505389599 (H.F. and P.K.); the Excellence Initiative by German Federal and State Governments (CECAD) (H.F. and M.T.); German Centre for Diabetes Research, project IDs 82DZD00502 and 82DZD05H1G (M.T.); DFG through SFB 1451, project ID 431549029, C06 (M.T.); the Mapping Autonomic Neural Interaction and Control

(MANIAC) Emerging Group by the University of Cologne Excellent Research Support Program (M.T.); and the Israel Science Foundation, grant 3015/22 (Y.L.). **Author contributions:** Conceptualization: M.M., H.F.; Funding acquisition: M.T., Y.L., H.F.; Investigation: M.M., H.W., B.Ku., A.C., S.L., D.F.; Methodology: M.M., D.F., L.T., S.S., I.T., Y.P., J.D., B.Ku., M.T., B.Ki., Y.L., H.F.; Project administration: M.M.; Supervision: P.K., M.T., Y.L., H.F.; Visualization: M.M., H.F.; Writing – original draft: M.M., H.F.; Writing – review & editing: M.M., B.Ku., J.D., S.L., B.Ki., M.T., Y.L., H.F., with input from all authors. **Competing interests:** S.L. and J.D. are Novo Nordisk employees and minor shareholders as part of an employee offering program. The authors declare no other competing interests. **Data and materials availability:** All data are available in the main text or the supplementary materials. In-house analysis scripts for fiber photometry are available at Zenodo (*56*). The human data reported in this study cannot be deposited in a public repository per General Data Protection Regulation and Institutional Review Board data protection policies. To request access, please contact the senior author (H.F.). Data provision may include processed and unprocessed data and will require a data sharing agreement. Data sharing necessitates that the purpose of data reanalysis is in line with the study aims as approved by the ethics review boards and participant consent. Furthermore, consent to data privacy must be assured by signing the agreement form accordingly. Requests will be answered within 4 weeks.

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adp1510
Materials and Methods
Figs. S1 to S9
References (*57*–*60*)
MDAR Reproducibility Checklist

Submitted 10 March 2024; accepted 11 December 2024
10.1126/science.adp1510